Advanced Properties of Threshold Cryptosystems

Graduate Internship (Master 2), Summer 2027

Location	PQShield SAS, Paris, France 8 Rue des Pirogues de Bercy
Contact	Guilhem Niot, Thomas Prest
Starting date, duration	Q1-Q2 2027 (flexible), 6 months
Gross salary	1350 € / month

Internship

Background

In a threshold scheme, the secret key is split between several parties, and a minimum number of them must work together to sign or decrypt. An attacker who corrupts fewer parties than this threshold learns nothing useful about the key. These schemes are already used in practice, for instance to protect digital assets. In 2025, NIST started the Multi-Party Threshold Cryptography (MPTC) project and opened a call for threshold schemes, with post-quantum schemes in scope.

PQShield researchers have co-designed several lattice-based threshold signature and encryption schemes that take part in this process, and already provide competitive efficiency. Deployed systems, however, rely on additional features that go beyond the core threshold signing or decrypting, and for post-quantum schemes these features are either missing or expensive. Some of the building blocks used in these schemes also leave room for improvement.

The goal of this internship is to add some of these advanced features to post-quantum threshold schemes, and to make these schemes more efficient.

Objectives

The intern will first study the existing post-quantum threshold schemes and the features that applications expect from them. They will then design new constructions or optimizations for some of these schemes and prove their security. Depending on progress,

the work can include an implementation and benchmarks, and may lead to a publication or feed into PQShield's contributions to the NIST MPTC process.

About PQShield

PQShield is a cybersecurity scaleup that specialises in post-quantum cryptography, protecting information from today's attacks while preparing organisations for the threat landscape of tomorrow. It demonstrates quantum-safe cryptography on chips, in applications and in the cloud. We are headquartered in Oxford, with additional teams in the Netherlands, Germany and France.

PQShield SAS, based in Paris (France), concentrates the research activities of PQShield. Our mission is to come up with innovative algorithmic and/or protocol-level solutions to real-world cryptographic problems. Besides post-quantum cryptographic primitives, our research interests include implementation security and advanced cryptosystems and protocols such as secure messaging, threshold schemes, and multiparty computation.

Requirements and how to apply

Ideal candidates should possess as many of these qualities as possible:

- Familiarity with lattice-based cryptography.
- Interest in provable security and cryptographic protocols.
- Knowledge of secret sharing, multi-party computation or zero-knowledge proofs is a plus.
- Programming skills (C, Rust, Python) are a plus.
- Strong autonomy.
- Good communication skills.

Supervision: The intern will be supervised by Guilhem Niot (Senior Cryptography Researcher) and Thomas Prest (Lead Cryptography Researcher), who have co-designed several lattice-based threshold schemes, including candidates in the NIST MPTC process.

References

1. NIST, *NIST First Call for Multi-Party Threshold Schemes*, NIST IR 8214C, 2026. https://doi.org/10.6028/NIST.IR.8214C
2. S. Celi, R. del Pino, T. Espitau, G. Niot, T. Prest, *Efficient Threshold ML-DSA*, USENIX Security 2026. https://eprint.iacr.org/2026/013

3. G. Borin, S. Celi, R. del Pino, T. Espitau, S. Katsumata, G. Niot, T. Prest, K. Takemure, *Hermine: An Efficient Lattice-based FROST-like Threshold Signature*, ASIACRYPT 2026. https://eprint.iacr.org/2026/419
4. K. Boudgoust, O. Lapiha, R. del Pino, T. Prest, *IND-CCA Lattice Threshold KEM under 30 KiB*, PKC 2026. https://eprint.iacr.org/2026/021